



Path Detection in Virtual Environment for Synchronous EEG by Density Based Support Vector Machine

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Abstract: The use of Brain-Computer Interface (BCI) has been increasing exponentially in the recent years due to the use of low-cost commercial Fast Fourier Transform (FFT) based EEG reading devices with nonclinical accuracy for consumer application development. Also, the design and implementation of 3D virtual environments for BCI training purposes has proven to be effective due to the high interaction with the end user and the assistance for recreating a specific type of signal or behavior. The aim of this paper is to present a method and the results of applying a binary Density Based Support Vector Machine (DBSVM) Classifier in a 3D virtual environment designed for interaction with EEG predefined signal patterns. The environment trains the classifier by taking 180 second EPOCHs and classifying them into a successful/unsuccessful attempt per test subject. The applications can be extended for implementing mind-wave pattern password or tracing a specific set of mind-based commands for virtual path tracing purposes. The tested SVM had a success rate of 60%. Further work includes the study of different classifier features and implementation of a dynamic classifier.

Keywords: BMI, MindWave, SVM, Pattern Recognition, Machine Learning

I. BASIC FFT-BCI BACKGROUND

The use of FFT-BCI based 3D virtual environments has been implemented effectively in different application fields; it is rather a new technological tool, which may be exploited to enhance motor retraining [1]. A simple development with software like Unity3D can be combined with other assistance technologies [2] or interfaces [3] in order to improve a specific system's performance with an end user's feedback. The use of these interfaces are either interacting with the preprocessed behavior signal output data stream from a processing IC with the aid of "slow adaptive algorithms" [4] like the TGAM1, or the RAW-EEG value obtained from an active electrode and then decomposed into a specific behavior according to the spectral density of the sample [5]. These waves, preferred range for processing are specified in Table 1 [6]. There are algorithms that focus on detecting a specific emotion from the end user according to their mind-wave patterns. Some of them can map and help to differentiate emotions like attention/inattention and meditation/uneasiness based on a simple beta/theta wave ratio as considered by Equation (1) [7], where *SMR* stands for low Beta waves (12-15 Hz) and *mBeta* for middle-Beta waves (16-20 Hz).

Table.1 Basic human brain waves and ranges

Name	Frequency range [Hz]	Preferred range [Hz]
Delta [δ]	0.5-2.75	0.5-3.0
Theta [θ]	3.5-6.75	3.0-7.0 or 4.0-8.0
Low alpha [$l-\alpha$]	7.5-9.25	8.0-10.0
High alpha [$h-\alpha$]	10.0-11.75	10.0-12.0
Low Beta [$l-\beta$]	13-16.75	SMR 12.0-15.0 Low Beta 15.0-18.0
High Beta [$h-\beta$]	18-29.75	High Beta/ 23.0-38.0 Low Gamma
Low Gamma [$l-\gamma$]	31-39.75	Mid Gamma
Mid Gamma [$m-\gamma$]	41-49.75	High Gamma

$$Emotion_{ratio} = \frac{SMR+mBeta}{Theta} \quad (1)$$

This paper describes the implementation and results of a Density Based Support Vector Machine Classifier from a series of training data sets obtained from a 3D environment. The classifier's input data are a binary set of 180 long signal vectors (EPOCHs) corresponding to the preprocessed attention and meditation outputs from a one dry electrode EEG headset with the configuration as shown in Figure 1. An Arduino Uno board was used

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as an RF/RS232 data bridge between the headset and the computer script and an Arduino Micro was used for keyboard simulation purposes.

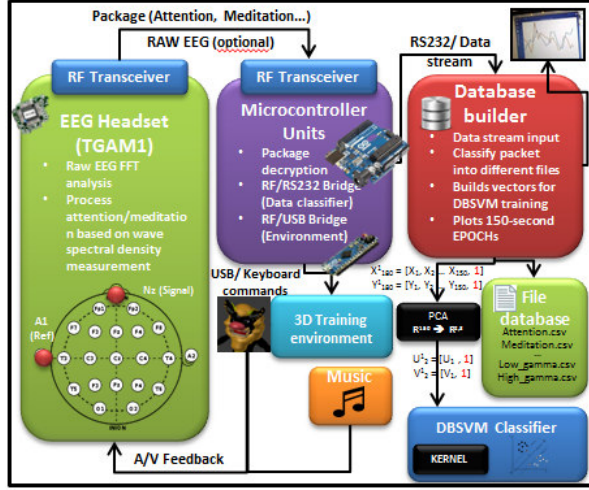


Fig. 1 System description

Previously, the user performance for a specific signal pattern with a single electrode EEG commercial tool proved to be reliable if good timing is given between attention and meditation intervals [8], especially for attention and meditation patterns as shown in Figure 2.

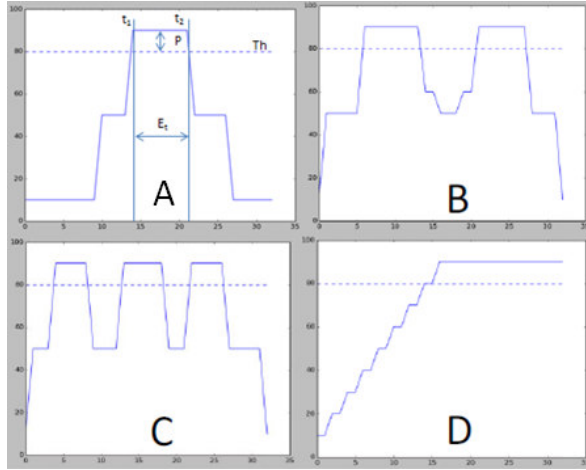


Fig. 2 Attention/Meditation example patterns

The aim of this research is to elaborate a method for recognizing mind behavior patterns for a password application development.

II. DENSITY BASED SUPPORT VECTOR MACHINES

Machine Learning techniques are generally helpful in feature extraction among EEG signals, since the recorded signals usually tend to have variable bias and uncertain changes, which make the feature classification difficult. The Support Vector Machine (SVM) model proposed by Dr. Vapnik has been used as a form of supervised learning models (SLM) in detecting certain features in the captured EEG data and in many other systems [9]. However, the sensitivity of SVM to outliers is a weak point for this algorithm. One of the approaches for reducing this sensitivity is the Density Based Support Vector Machine (DBSVM) method, which is based in the simple expression of population density according to Equation (2) [10]:

$$\text{Population density} = \frac{\text{number of population}}{\text{area}} \quad (2)$$

Although the concept of population density is used to develop the DBSVM model, the formula is actually based on two separate algorithms to remove the outliers based on a vector's total Euclidean or Mahalanobis distance between its averages [11].

A. DBSVM with Euclidean distance

For instance, let's suppose a set of n numbers of two-dimensional data $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ is given. The Euclidean distance between two points is given by Equation (3):

$$D(1,2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \quad (3)$$

The next step is to sum up all the distances for each point (i.e. $d_1 = [D(1,2) + D(1,3) + \dots + D(1,n)]$) where n is the number of data points in the data set. The total distance for all data points of one data set is needed to calculate the average distance that will be used to determine data points inside and outside of densely populated area. Equation (4) shows the expression to calculate this:

$$\text{Average}_d = \frac{\sum_{j=1}^N \sum_{i=1}^N \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}}{n} \quad (4)$$

$$\text{if } d_i > \text{Average}_d \rightarrow x_i = \text{outlier}$$

After calculating $Average_d$ by formula, those points that are not considered as outliers will become the new training data set in which the points will be considered important data points.

B. DBSVM with Mahalanobis Distance

The Mahalanobis distance is the distance between the x from the quantity μ . This distance is based on the correlation between variables or the variance-covariance matrix S . To determine the distance from two data points, the formula from Equation (5) is used:

$$D_m(x) = \sqrt{(x - y)^T S^{-1} (x - y)} \quad (5)$$

Therefore, as used for the Euclidean Distance average calculation, the average Mahalanobis distance for excluding outliers is as described in Equation (6):

$$Average_d = \frac{\sum_{i=1}^N \sqrt{(x_i - y)^T S^{-1} (x_i - y)}}{n} \quad (6)$$

The data sets obtained from the training environment will be tested with both population density methods for success rate comparison. The application of this method can help for having a maximized margin and a better generalization of the pattern EPOCHs obtained from the training.

III. THE 3D TRAINING ENVIRONMENT

The use of simulators for training with EEG Biofeedback is a common application and has been proven effectiveness in solving problems such as attention enhancement [12]. The proposed training environment was designed with the use of Blender for the animated models and structures as well as Unity3D for its ease of use and deployment. The environment has a series of features such as music and interaction with the experiment variables that are described as follows. The environment setup was created for the test subjects to familiarize with the objects and understand the way of interacting with them; an additional relaxation music track can be used for aiming in obtaining higher meditation behaviors. The purpose of the training task is to guide the player (training subject) throughout the predefined path, as specified in Figure 3. When concentrated, the user can move the player forward and then move the player to the right if the user begins to enter meditation. In either case, the user can stop the player by blinking the eyes constantly until the player stops from produced artifact noise.

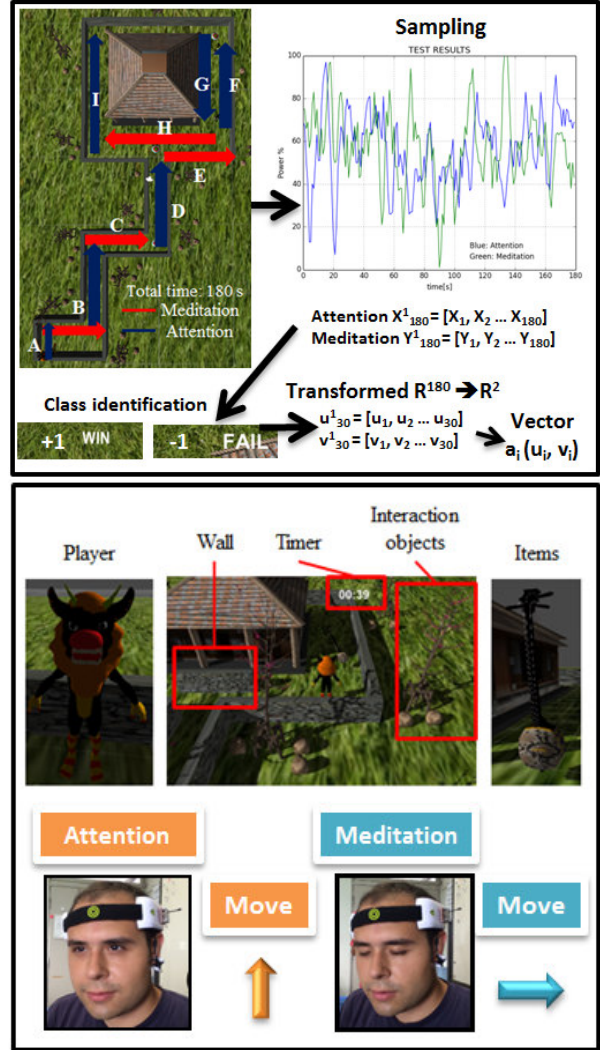


Fig. 3 Environment description

If the mind-wave series came into the expected pattern in the required time, the attempt will be classified as successful and the class $+1$ will be labeled for the output EPOCH. However, if the test subject is not able to perform the entire track along the timeframe, an unsuccessful attempt is recorded with the -1 label. The resulting vectors will be used as samples for later mapping in the SVM classifier. Brain waves are dependent on each test subject, and so the performed wave patterns and headset calibration. Due to this factor, the training requires to be performed individually. Fortunately, the device has a fast self-calibration routine [13] and a trained subject can easily adapt in controlling the headset in minutes. In order to have a

good classifier, 30 samples were obtained from the test subject. The testing area was a quiet room with the test subject sitting in front of a screen displaying the 3D environment and the data processing script running in the background for obtaining the corresponding data sets. Whenever the test subject was ready to perform a test a red button was pressed to restart the environment and script.

IV. DBSVM CLASSIFIERS AND RESULTS

Once the test was performed, as there are two outputs from each test, matrix A consisting of attention values and matrix B with meditation values. Feature X from Equation (7) was empirically obtained and used to relate both matrices (by component calculation for the end feature vector for best separation):

$$X = \varphi \left[\frac{A^2(A-B)}{A+B} \right] ; \varphi = 0.001 \quad (7)$$

Where φ stands as a map resizing constant and the multiplication stands for element multiplication (if $A_i = -B_i$, $X_i = 0$). Once the matrices have been associated with the feature, the PCA matrix ϕ is obtained for matrix X . The two main components the transform matrix Y detailed by Equation (8) were used to map the outputs in the two-dimensional space.

$$X\phi = Y \quad (8)$$

For instance, the first component of the first row of the matrix will be the x coordinate and the second component will be the y coordinate. For this DBSVM test, a basic hard margin was established by observation from the nearest supporting vectors and the possible limits for the function. Once the DBSVM has been set, there was a test vector consisting of five test elements X_i with their respective label. This time the mapping coordinates were obtained by using the PCA matrix ϕ described in (8). The corresponding classification success chart for different kernels is described on Table 2 and a resulting map for Euclidean DBSVM is shown in Figure 4.

Table.2 Classification Error Rates

Kernel	Normal	Euclidean	Mahalanobis
Linear	20%	60%	60%
RBF	40%	40%	40%
Polynomial	60%	20%	20%

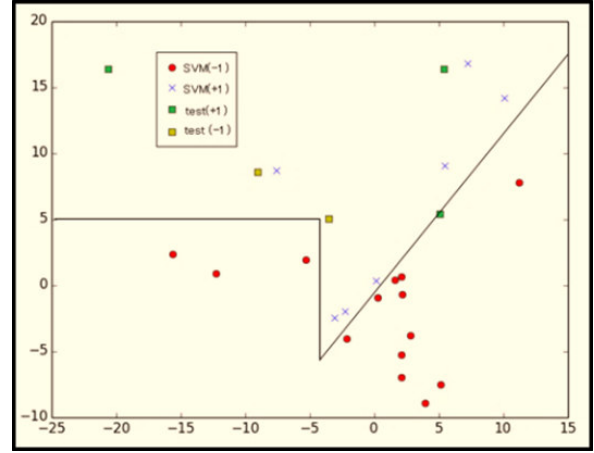


Fig. 4 DBSVM by Euclidean with test data set

V. CONCLUSIONS AND DISCUSSION

This paper presents a way to use the Density Based Support Vector Machine for a synchronous mind pattern classification problem. One of the main disadvantages of the SVMs is the selection of a particular feature that can reduce the classification error. By choosing the correct feature, datasets can be condensed in specific regions and clusters in which the Euclidean and Mahalanobis outlier discrimination algorithm can perform as expected. For this particular problem, the selected feature was the one that worked best after trying various models with the use of PCA. There can be other mapping approaches that can be performed based on the Attention/Meditation signal correlation. However, the use of the raw EEG data sets could represent a better way for classification derived from training. The performed test demonstrates the effective use of low-cost, single electrode commercial FFT-based EEG headsets for simple classification tasks with the aid of a simple training environment with optional music feedback. Nonetheless, it is important to choose a specific feature to reduce the error. Other linear kernels have been tested for the previous dataset with a better recognition rate [14], however the use of a three dimensional classifier is also in discussion.

The use of more data is important to enhance the supervised learning model, for this purpose, the use of a dynamic, self-adapting classifier could improve the recognition rate, e.g. a fifty training set that can perform multipoint search and throw the least convenient set after one test has been classified successfully and added for the next pattern recognition sample.

VI. FURTHER WORK

From the DBSVM, a more specific feature will be determined based on the Attention/Meditation wave pattern behaviors from the obtained EPOCHs. Also, the use of a parallel testing for multipoint search and a dynamic classifier is in discussion to obtain a more precise classifier that can be adjusted for a certain number of training samples.

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